

Visual Simulation of Weathering By γ -ton Tracing

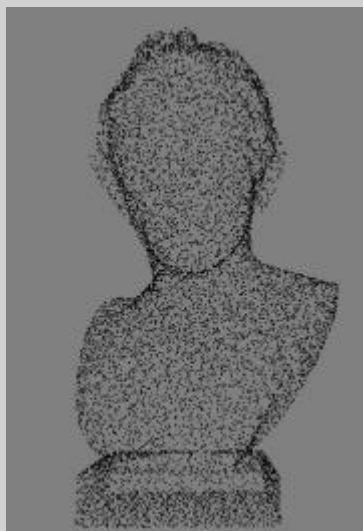
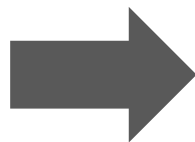
2020105671
컴퓨터공학과
하대진

Input

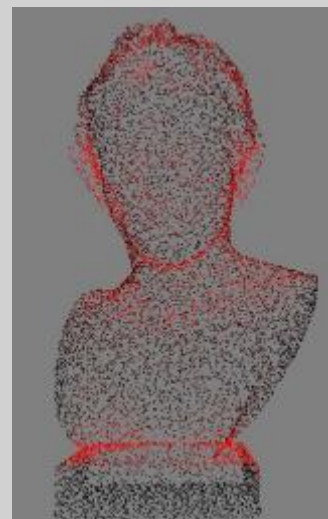


Mesh

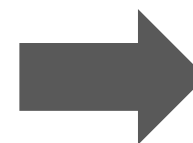
+ 샘플 텍스처
(녹, 이끼 등)



Resampled
Surfel



Γ -map

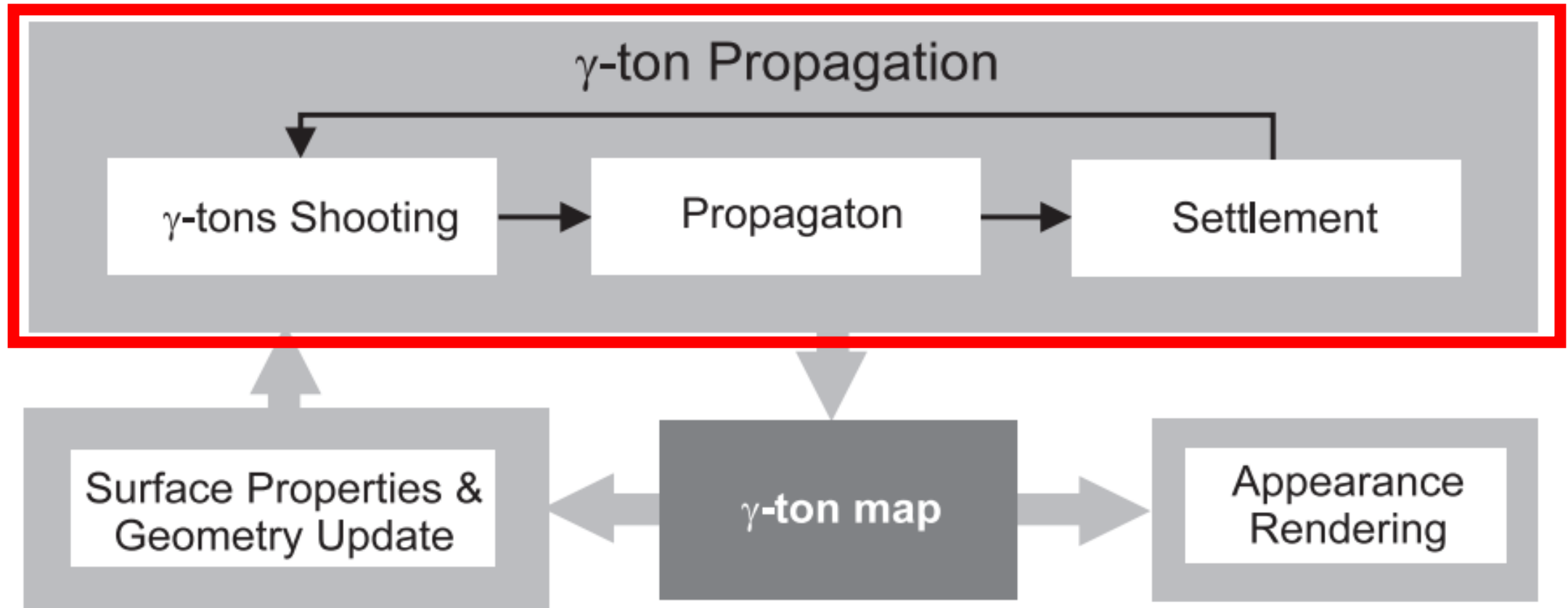


Output



Texture Mesh

Pipeline



Propagation

감마톤 입자는 두 가지 주요 속성을 가집니다

- Motion Probabilities (입자의 운동 상태)
- Carrier Attributes (입자의 특성)

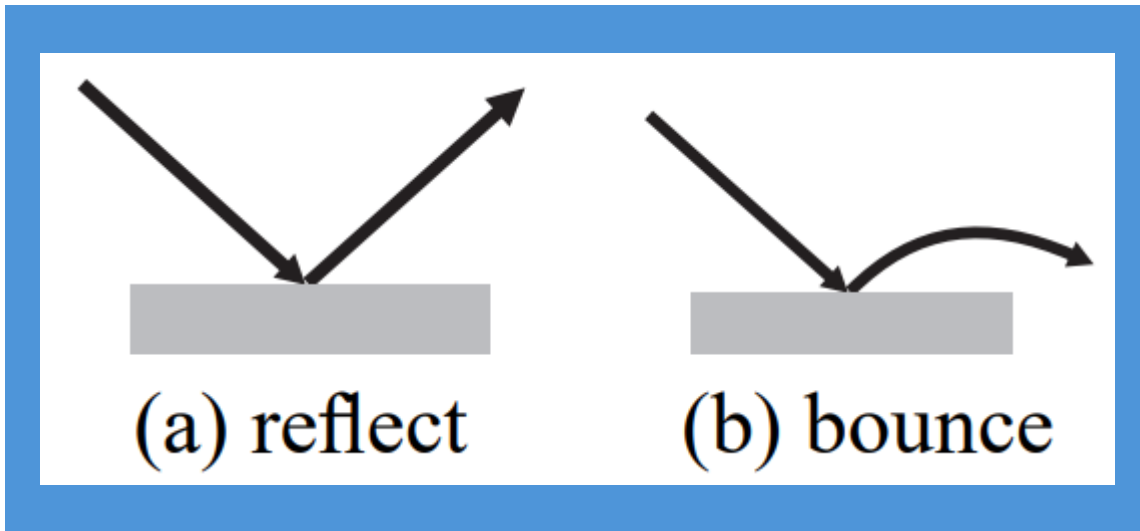
Propagation

```
Ton {  
    Motion probabilities:  
         $k_s$ ,  $k_p$ ,  $k_f$ ; // Probability of propagating along a straight line,  
                        // parabolic curve, or flowing  
    Carrier attributes:  
         $s_h$ ; // Humidity  
         $s_d$ ; // Amount of dirt  
         $s_f$ ; // Amount of fungus  
        ...  
}
```

Single Layer로 제안되어 있기에, 구조체 구현에 용이하다.

Motion Probabilities

Diffusion과 동일하게 구현
evenly distribute over the upper hemisphere

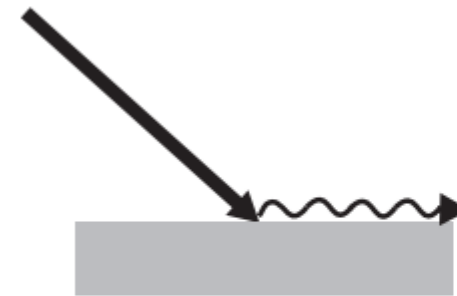


k_s

Ray 이론과 동일

k_p

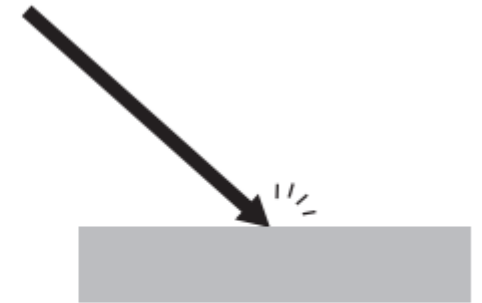
포물선 운동



(c) flow

k_f

표면에 붙어
step만큼 이동



(d) settled

$$1 - k_s - k_p - k_f$$

Motion Probabilities

Uniform Sampling of Hemisphere

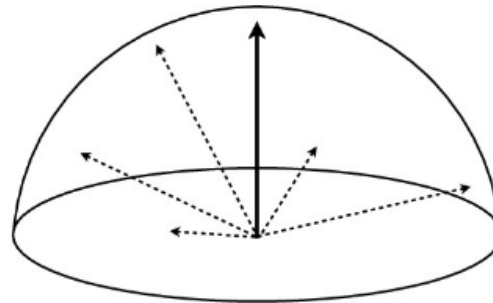
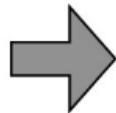
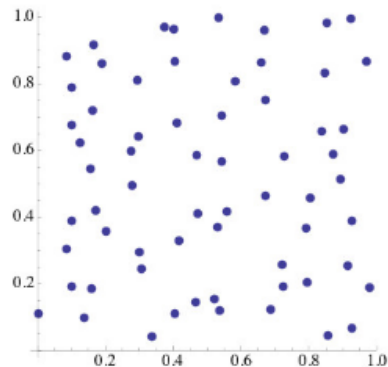
Generate random direction on hemisphere (all dirs. equally likely)

$$p(\omega) = \frac{1}{2\pi}$$

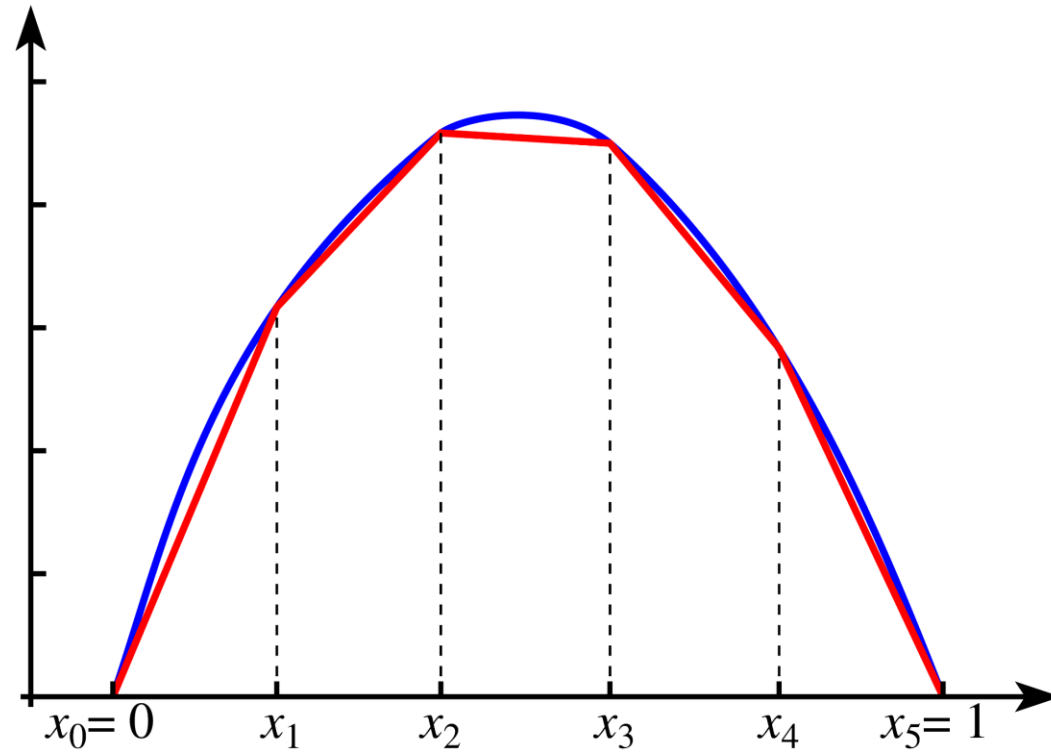
Direction computed from uniformly distributed point on 2D square:

$$(\xi_1, \xi_2) \rightarrow (\sqrt{1 - \xi_1^2} \cos(2\pi\xi_2), \sqrt{1 - \xi_1^2} \sin(2\pi\xi_2), \xi_1)$$

Q. 왜 램버시안 코사인 법칙을 사용하지 않았나?



Piecewise Linear



Bounce에 있는 포물선 궤적을 Piecewise Linear로 근사

For a γ -ton traveling along a parabolic curve, its trajectory is piecewise linearly approximated

r-Reflectance & Material Property

감마톤 입자에 대응해서, 각 Surfel은 두 정보를 가집니다.

- r-reflectance (감마톤이 충돌할 때, k를 얼마나 깎아낼지)
- Material Property (표면에 쌓인 풍화 물질의 양)

r-Reflectance & Material Property

```
Surf {  
     $\gamma$ -reflectance:  
         $\Delta_s, \Delta_p, \Delta_f;$     // Deterioration rate for  $k_s, k_p,$  &  $k_f$   
    Material properties:  
         $s_h;$                       // Humidity  
         $s_d;$                       // Amount of dirt  
         $s_f;$                       // Amount of fungus  
        ...  
}
```

r-Reflectance & Material Property

$$k'_s = \max(k_s - \Delta_s, 0)$$

$$k'_p = \max(k_p - \Delta_p, 0)$$

$$k'_f = \max(k_f + \max(k_p - \Delta_p, 0) - \Delta_f, 0)$$

적용

1-1. 감마톤 초기값

$$k_s = 0.0, \quad k_p = 0.8, \quad k_f = 0.2$$

$$s_{\text{humidity}} = 1, \quad s_{\text{rust}} = 0$$

1-2. 계단, 체인의 반사율

$$\Delta_s = 0.0, \quad \Delta_p = 0.4, \quad \Delta_f = 0.05$$

$$s_{\text{humidity}} = 0, \quad s_{\text{rust}} = 0$$



적용

2. 시멘트 계단/벽에 입자가 닿을 때

$$\text{Surf.}s_{\text{humidity}} \leftarrow \text{Surf.}s_{\text{humidity}} + 0.1 \cdot \text{Ton.}s_{\text{humidity}}$$

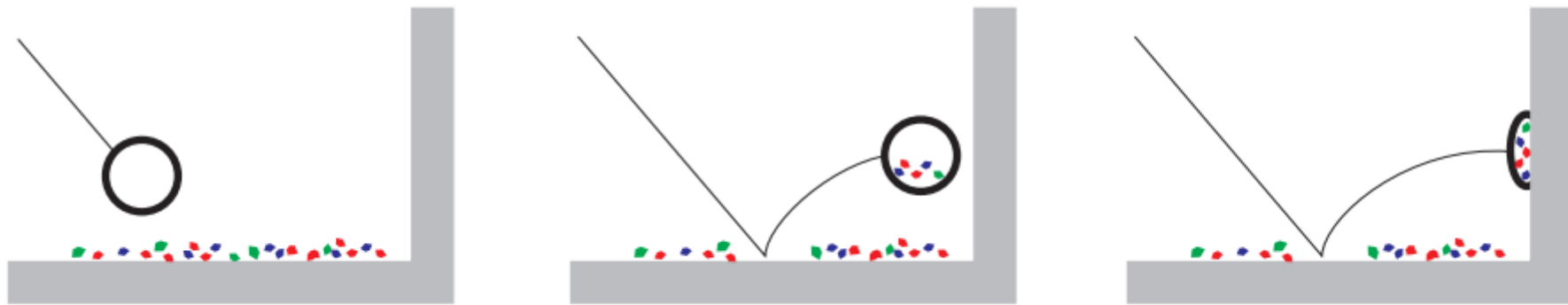
$$\text{Surf.}s_{\text{rust}} \leftarrow \text{Surf.}s_{\text{rust}} + 0.1 \cdot \text{Ton.}s_{\text{rust}}$$

3. 금속 체인에 입자가 닿을 때

$$\text{Surf.}s_{\text{humidity}} \leftarrow \text{Surf.}s_{\text{humidity}} + 0.1 \cdot \text{Ton.}s_{\text{humidity}}$$

적용

4. 입자가 체인에서 녹을 집어들 때



$$\text{Ton.}s_{\text{rust}} \leftarrow \text{Ton.}s_{\text{rust}} + \text{Surf.}s_{\text{rust}}$$

적용

5. 표면 속성 업데이트

$$\text{Surf.}s_{\text{rust}} \leftarrow \text{Surf.}s_{\text{rust}} + 0.015 \cdot \text{Surf.}s_{\text{humidity}}$$

$$\text{Surf.}s_{\text{humidity}} \leftarrow \text{Surf.}s_{\text{humidity}} - 0.5 \cdot \text{Surf.}s_{\text{humidity}}$$

적용



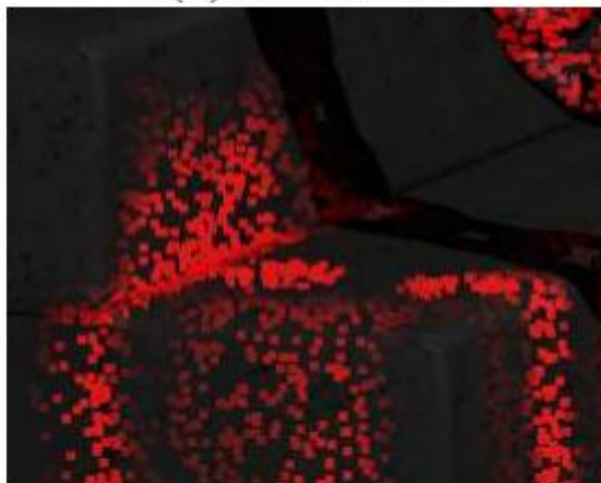
(a) original



(b) weathered

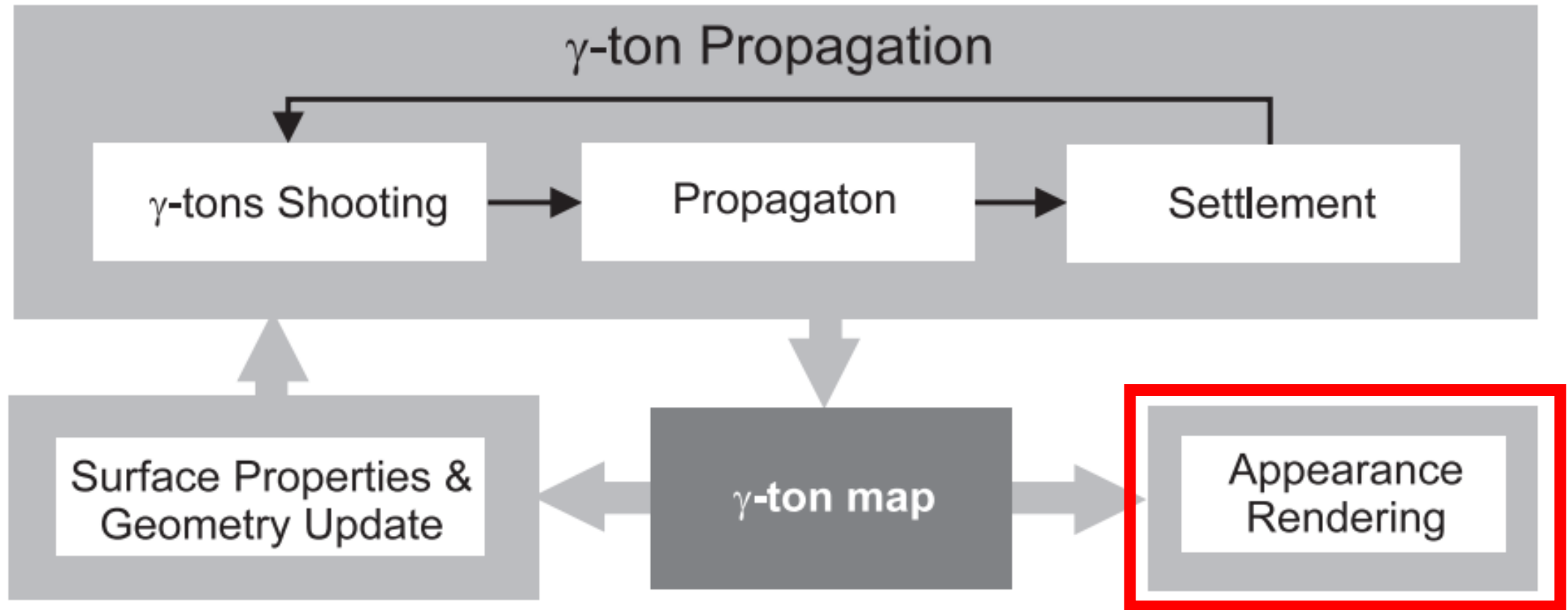


(c) blow-up



(d) γ -ton map

Pipeline



Multi-Texturing



$$C_{final} = (1 - s) \cdot C_{base} + s \cdot C_{weathered}$$

r-ton map에서 추출한 지점에서 Alpha-blending (선형 보간)을 사용
(PBR의 영향을 미치지 못함)

Displacement-mapping



Figure 11: Erosion of Sphinx after thousands of years.

$$P_{new} = P_{old} + (k \cdot \Delta M) \cdot N$$

r-ton map에 기록한
Delta-M에 따라 변화